

Routine Data Quality Audit Toolkit: Overview and User Guide



Adapted from Zambia's HMIS Data Quality Audit Guidelines (2014)

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General Overview

Every data collection system is prone to errors resulting from design oversights or human errors during data collection, processing, and transmission. Before data are used for decision-making, all errors should be removed (data processing). Data errors that occur during the data collection stage are the most difficult to resolve. Therefore, it is important that errors are minimized as much as possible, starting with individual patient data. This is only achievable through adequate training and consistent use of standardized guidelines at each stage.

Objectives of Conducting a Data Quality Audit

When used routinely, a data quality audit is one of the best measures used to improve data integrity once data has been collected and before it is used for important decisions. The objectives of conducting a data quality audit are:

1. To quickly confirm the validity of data reported by service delivery points and/or by district health teams.
2. To identify the determinants of data quality.
3. To use the results from the audit to develop action plans to improve data quality (and indirectly, quality of services).
4. To correct inaccurate data that has been transmitted to the relevant reporting system (e.g., HMIS DHIS2), when feasible.
5. To provide a process and output of data quality that can be compared across time and place as a way of monitoring the performance of the HMIS.

The Data Quality Audit procedures and tools in this toolkit are meant to:

1. Assess the quality of reported data.
2. Review the health information system processes that produce the data.
3. Develop action plans to improve data management and quality.

Data Quality Definitions

In the context of data quality audits, data quality is defined through the following dimensions:

- Reporting completeness: Percentage of reporting periods for which data has been audited that data was reported into the HMIS (Percentage of expected monthly reports submitted)

- Data element completeness: Percentage of reporting periods for which there were no data elements missing in submitted reports (Number of monthly reports with 100% of data elements submitted/number of expected monthly reports)
- Reporting timeliness: Percentage of reporting periods for which data was reported into the HMIS by the due date (number of monthly reports submitted on time/number of expected monthly reports)
- Accuracy (of data): Percentage of all data elements reported accurately into the HMIS (number of periods of accurately reported data elements/number of periods of reported data elements)
- Accuracy (of reporting periods): Percentage of reporting periods for which all data elements were reported accurately into the HMIS (number of accurately reported data elements/number of reported data elements)

Contents of Data Quality Audit Toolkit

- DQA Overview (this document)
- Module 1 Data Quality Audit Template.xlsx
- Module 2 DQA Data Preparation Tool.xlsm
- Module 3 Data Quality Audit Results Data Prep.xlsm
- Tally sheet sample.xlsx
- Facility performance assessment questionnaire sample.docx
- RDQA Orientation agenda sample.docx
- Data Quality Audit Report template outline.docx

Audit Approaches

There are two approaches that can be considered when planning an audit—sequential or standalone:

1. **Single-level (Stand-alone):** This approach is when the district health office decides to undertake an audit of selected facilities or data elements as a local level initiative, supported by themselves or through a partner. This can either be a planned activity or an ad hoc one. Ad hoc audits include those activities undertaken because a problem has been identified and an audit is ordered. HMIS reporting from the service delivery point is typically assessed alone.
2. **Dual-level (Sequential):** This approach applies to instances whereby an administrative level higher than the district (e.g. provincial or national) commissions an audit throughout the country or in a given province or a district. In such a case, the audit should start with the district health office and then onto health facilities of interest. HMIS reporting from both the service delivery point (source reporting) and the district (secondary reporting) will be assessed.

Preparing the audit

Preparations are done at the level commissioning the audit (district, province, national). The selection of data elements and facilities to be included in the audit should be transparent and with input from stakeholders relevant to the data and geographies being audited. The preparatory stage is made up of the following steps:

1. Selecting data elements

One audit is not able to cover all data elements from the service delivery and disease forms. Therefore, it is necessary to select data elements before the audit. One approach may be to

select data elements that the planning team consider to be most impactful to decision-making. Another approach may be to take a random selection of data elements, with each data element given an equal chance of being included.

Additionally, decide whether the audit has the resources and need to include administering a facility performance assessment to determine if data collection and reporting procedures are understood and being applied correctly. If so, refer to the sample performance assessment questionnaire and revise as appropriate to the local context.

2. Selecting data audit sites

- **Choosing sites:** Although all facilities are eligible for a data audit, it is usually not possible to cover all facilities in one round of an audit. The level commissioning the audit should select sites for each round of the audit.
 - **Note:** If audits are conducted routinely over the course of a year, then each facility should have undergone at least one audit annually. If audits are conducted annually, then each facility should receive an audit at least once every two years.
- In choosing health facilities to include in the audit, the audit team may explore different factors to select sites, including random selection. Audit teams may also consider the following aspects:
 - Conduct a preliminary assessment of completeness and timeliness of reported data for all sites within the period under review. The audit team may consider prioritizing those sites that appear to be struggling so that a detailed review can be done through the audit to inform key areas for improvement.
 - Review previous audit results and associated data improvement action plans. The level commissioning the audit may choose to follow up on those sites that should have implemented remedial measures to assess progress.

3. Selecting the data quality audit team

Before an audit can take place, the level commissioning the audit will need to determine the team composition. This will vary depending on the specific objectives of the audit and the level that is organizing the audit. Team members to consider include Health Information Officers (subnational or national levels), service-specific health officers (e.g., Malaria Focal Points), other medical or planning officers (subnational or national). Another consideration is to include team members from other districts to ensure objectivity and facilitate exchange of experiences.

4. Review relevant documentation

Documents that should be reviewed may include previous audit reports; HMIS feedback reports; most recent supervision report or performance assessment/technical support report; annual/quarterly action plans; and any relevant information hosted by individual program managers that may not be in the listed documents. Obtaining as much detail as possible about the targeted site may reduce the amount of time spent on site.

5. Prepare data for the audit

Once the data elements for the audit have been selected, the audit team should now prepare the data for audit. This **MUST** be done **BEFORE** the actual fieldwork commences.

- Module 1 in the RDQA toolkit, “Data Quality Audit Template”, must be configured according to the data elements that have been selected and the data source being audited. Instructions on how to configure the tool are contained within the tool.

- Module 2 in the toolkit, “DQA Data Preparation Tool”, will be used to prepare the audit workbooks for each targeted site. Instructions on how to use the tool are contained within the tool.

6. Orientation of data quality audit teams

Members of the audit team should receive an orientation on how to conduct the audit and use the data quality audit tools. The orientation should include a hands-on review of the audit tool and may include using dummy registers or photos of facility register pages to practice tallying and recording data in the audit tool. Orientation should also include a review of the definitions of data quality indicators included in the analysis outputs, discussion of the analysis outputs of the practice workbooks, and guidance on developing the follow-up action plans included in the DQA tool. For reference, data quality indicator definitions and guidance for reviewing analysis outputs and developing action plans are included in the DQA tool on the Audit Instructions worksheet.

A sample agenda for RDQA orientation is included in the toolkit.

7. **Prepare for site visits:** It is recommended that target health sites are informed in advance about the planned visit, preferably at least one month before visiting the site. Therefore, it is recommended that the notice to the sites includes the following information:

For audits organized and led from the central level, the District Medical Office should be informed of the following:

- The District Medical Officer should be present and guarantee access to their office.
- The officer in-charge of the HMIS and the DHIS should be present.
- Relevant program officers should be available for the audit.
- All HMIS forms submitted by health facilities must be available.
- Date and time the audit will take place.
- Anticipated duration of the audit.

The facility level should be informed of the following:

- The in-charge of the facility should be present, or delegate to relevant staff. For hospitals, the Director and the departmental or ward in-charges should be available for the audit.
- Representation from staff who complete registers, activity sheets, and tally sheets.
- Availability of staff that are responsible for preparing the HMIS reports used to enter data into DHIS2.
- Guaranteed access to patient cards, registers, activity sheets, and tally sheets.
- Date and time the audit will take place.
- Anticipated duration of the audit.

Fieldwork

District health office

At the district health office, secondary report data should be entered in the “Secondary Report” worksheet in each facility DQA file. On the “Source Data” worksheet, indicate whether data was submitted from the facility to the district office on time, according to guidelines on reporting timeliness.

Review the analysis comparing secondary reports with data in DHIS2. When the secondary reports have been used to enter data into DHIS2, any discrepancies found here are usually due to data entry errors.

Discuss procedures to prevent data entry errors.

Facility

At the facility, the audit team lead should meet with the facility in-charge and explain the mission, including the objectives of the audit and how the exercise will benefit the facility, the patients, and the country. Confirm whether the notice sent about this visit was received and if all the things requested have been prepared for the audit.

Request a space for the team to conduct the audit that won't disrupt the functioning of health facility activities.

It is a common occurrence that even if prior notice is given, health workers are always busy when visited. Do not disrupt normal patient care. If the registers and cards included in the review are currently in use, exclude those records until they are free.

Tally data from relevant registers for the selected data elements using the provided tally sheet, and then enter the data into the "Source Data" worksheet.

Once the data is entered in the DQA tool, ask the facility in-charge to gather the health facility staff who record, tally, and report data. With the health facility team, review reporting periods and data elements with errors, review data quality indicators, and discuss. Try to identify the causes for errors and discuss how errors can be minimized or eliminated in the future. Also consider whether correct data was entered for the wrong reporting period (e.g., there can be situations in which all data is shifted by one month – review the data in both the "System Data" and "Source Data" worksheets).

Based on the audit findings and discussion, develop an action plan with the health facility staff to improve the quality of data collected and reported. Include actions that:

- Health workers can take to avoid errors in recording data. For example, what information to include in diagnosis and treatment columns of registers.
- Health workers can take to avoid errors in compiling and reporting data. For example, reminders on what data to include for each data element, tips on filling out paper HMIS reporting forms.
- Province/District/NMCP/Partners can take to reorient or retrain facility staff in data recording and reporting.
- Province/District/NMCP/Partners can take to strengthen support and supervision at health facility to improve data recording and reporting skills and ensure quality of data.

If the facility has access to a computer, provide a copy of the DQA tool and action plan. Before leaving the facility, thank the facility staff for their time and the work they are doing.

Debrief or review meeting

After the facility-based audits have been completed, it is useful to hold a meeting at the district or provincial level to review the results of the audit. Participants should include the audit teams, district and/or provincial health teams, and the facility in-charges, if possible. This meeting can be used to present the findings from each facility, review the facility follow-up action plans, and review HMIS data quality dashboards if they exist (e.g., in DHIS2, Tableau, or other dashboard tools). A recommended output of this meeting is a district or provincial-level data quality improvement plan to address data quality problems identified during the audit and to support follow-up of facility action plans. The plan should identify the individuals responsible for implementing the plan, the timeline, and resources required to ensure completion.

Prepare the audit report

Suggested audit report outline:

- I. Executive Summary
- II. Introduction and Background
 - Purpose of the DQA
 - Background on related HMIS efforts
 - Data Elements and Reporting Periods – Rationale for selection
 - Audit Sites and Rationale for selection
 - Description of the data-collection and reporting system (related to the data elements audited)
- III. Assessment of Data Quality
 - Description of the audit process
 - Reporting Timeliness
 - Reporting Completeness
 - Data Accuracy
 - Performance tables and charts
- IV. Determinants of Data Quality
 - Description of process to review determinants of data quality
 - Discussion of findings: overall strengths and weaknesses
 - Description of common follow-up action plan items
- V. Conclusion and Recommendations

Develop Data Quality Improvement Plan

Identified problems may fall into the following possible categories:

Peripheral level interventions: Some problems can easily be fixed without the involvement of higher administrative levels. Examples may include streamlining the filing system, restocking facilities with the appropriate versions/quantities of materials, withdrawing old versions of tools from sites, etc. Some solutions may require short implementation timespans while others may require more time and planning. All solutions that demand time to implement should be included in either the district annual plan or revised quarterly plans.

Higher level interventions required: Other problems may not be resolved by the peripheral level alone. For example, shortages of stationery (e.g., registers, reporting forms) may require the intervention of the central administrative level. When such problems are identified, the normal channels of notifying the higher levels should be used. If the problem requires technical input other than funds, this should be reflected in the plans using existing planning procedures.

Correction of Data in Reporting System

Ideally data that has been found to be incorrectly entered into the reporting system (e.g., DHIS2) should be corrected. If the system is configured to allow for corrections, individuals with appropriate authorization, such as District Health Information Officers, can use the “Corrections to make” worksheet in completed Data Quality Audit Templates (Module 1).